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REPUBLIQUE DU CAMEROUN

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MINISTERE DE L'ENSEIGNEMENT
SUPERIEUR



ACIAI 2026 COMPETITION

TECHNICAL SPECIFICATION SHEET

Fiche Technique de la Solution

MboaTrust AI

*An AI-Powered Financial Trust and Identity Platform
for Informal Merchants in Cameroon*

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Version 1.0

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Table of Contents

1. Team Identification	3
2. Application Domain	3
3. Expected Value-Add (Plus-Value Escomptée)	4
4. Target Audience & Market Niche	5
4.1 Primary Target	5
4.2 Secondary Target	5
4.3 Future Target	5
4.4 Market Niche	5
5. Detailed Development Process	6
5.1 Phase 1: Research & Data Collection	6
5.2 Phase 2: Data Processing & Preprocessing	6
5.3 Phase 3: AI Model Development	6
5.4 Phase 4: System Architecture & Integration	7
5.5 Phase 5: User Interface Design	7
5.6 Phase 6: Testing & Evaluation	7
5.7 Phase 7: Deployment & Future Roadmap	7
6. Estimated Implementation Budget	8
6.1 AI & Machine Learning Development	8
6.2 Software Development	9
6.3 Infrastructure & Operations	10
6.4 Human Resources & Project Management	10
6.5 Consolidated Budget Summary	11

Table of Tables

Table 1 - Team Identification	3
Table 2 - Expected Value-Add Summary	4
Table 3 - AI Module Development Costs	8
Table 4 - Software Development Costs	9
Table 5 - Infrastructure & Operations Costs	10
Table 6 - Human Resources & Project Management Costs	10
Table 7 - Consolidated Budget Summary	11

1. Team Identification

Table 1: Team Identification

Field	Details
Team Members	FUL MARVEL TIOBOU NAVOU MOMO LEBON
Academic Level & Program	Level 4 in Bachelor of engineering in Computer Science and Engineering
Team Supervisor	Mr. FOMAZOU TCHINDA CYRILLE BRICE
Institution	Institut Universitaire de la Côte (IUC)
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2. Application Domain

MboaTrust AI operates at the intersection of Financial Technology (FinTech), Artificial Intelligence, and Financial Inclusion. The platform specifically targets the informal economy sector in Cameroon, where mobile money transactions (MTN Mobile Money, Orange Money) are widespread but lack proper verification and record-keeping infrastructure.

The solution falls under the following domains:

- **AI-Powered Fraud Detection in Mobile Payments** - Using OCR and machine learning to verify payment screenshots and detect fraudulent transactions.
- **Digital Financial Identity for the Unbanked** - Building trust scores from transaction behavior to enable access to microfinance.
- **Small Business Financial Management** - Providing structured sales, expense, and profit tracking for informal merchants.

3. Expected Value-Add (Plus-Value Escomptée)

MboaTrust AI delivers significant value across multiple dimensions of the informal economy ecosystem:

Table 2: Expected Value-Add Summary

Dimension	Value Delivered
Fraud Reduction	Reduces financial losses caused by fake payment screenshots and deceptive transaction claims through AI-powered verification (OCR extraction + fraud pattern detection).
Financial Inclusion	Creates digital financial identities (Merchant Trust Scores) for previously “invisible” merchants, enabling them to demonstrate creditworthiness to microfinance institutions.
Business Intelligence	Transforms unstructured, informal financial activity into organized records with daily/weekly/monthly summaries of sales, expenses, and estimated profit.
Time & Efficiency	Automates payment verification that currently requires manual checking, saving merchants valuable time during busy market hours.
Trust in Digital Payments	Strengthens confidence in mobile money transactions, encouraging broader adoption of digital payments in informal markets.
Scalability	Although designed for Cameroon, the platform model is replicable across other African countries where mobile money and informal commerce dominate.

4. Target Audience & Market Niche

4.1 Primary Target

Informal merchants in Cameroon, specifically market traders (known locally as “bayam-sellam”), small shop owners, and street vendors who rely heavily on mobile money (MTN Mobile Money, Orange Money) for daily business transactions. Key characteristics include limited technical literacy, high reliance on mobile phones, frequent use of mobile money, preference for simple and fast interfaces, and time-sensitive market environments.

4.2 Secondary Target

System administrators and developers responsible for monitoring AI model performance, reviewing flagged suspicious transactions, and maintaining platform integrity.

4.3 Future Target

Microfinance institutions and digital lenders interested in leveraging Merchant Trust Scores for credit evaluation of informal-sector borrowers.

4.4 Market Niche

The solution occupies a unique niche at the intersection of AI-powered payment verification and financial identity generation for the unbanked informal sector. Unlike traditional banking tools or generic fintech apps, MboaTrust AI is purpose-built for merchants who operate without formal financial records and are excluded from conventional credit systems.

5. Detailed Development Process

5.1 Phase 1: Research & Data Collection

The project begins with a thorough analysis of the challenges facing informal merchants in Cameroon regarding mobile money fraud and the absence of financial records. Since real mobile money datasets are not publicly accessible, the team uses simulated datasets and publicly available data resembling mobile money transaction messages and payment proof screenshots. Synthetic transaction data is also generated for training anomaly detection models.

5.2 Phase 2: Data Processing & Preprocessing

All collected data undergoes rigorous preprocessing steps including text cleaning, normalization, feature extraction from transaction records, and image preprocessing for screenshot analysis. Image data is prepared for OCR processing, while structured text data is cleaned and formatted for machine learning model training.

5.3 Phase 3: AI Model Development

Multiple machine learning components are developed in parallel:

1. **OCR Module:** Extracts transaction amount, sender information, timestamp, and transaction reference from uploaded payment screenshots.
2. **Fraud Detection Module:** Analyzes extracted data for suspicious patterns including inconsistent structures, unusual formatting, altered screenshots, and mismatched values. Outputs a risk category, confidence level, and short explanation.
3. **Transaction Anomaly Detection Module:** Learns merchant transaction baselines over time and identifies deviations such as unusually large transfers or irregular payment behavior.
4. **Financial Behavior Analysis Module:** Evaluates daily sales patterns, transaction frequency, revenue consistency, and expense patterns to generate financial summaries and actionable insights.
5. **Merchant Trust Score Module:** Computes a composite score based on sales consistency, transaction frequency, expense behavior, and suspicious payment history. Presented with interpretive labels (low, moderate, strong).

5.4 Phase 4: System Architecture & Integration

The system is built as a three-tier architecture: a mobile-first Android application as the primary interface, a cloud-based backend server handling data storage, processing, and API communication, and AI processing modules operating as a dedicated service layer. Communication uses HTTPS with JSON data exchange. The system supports limited offline usage with automatic data synchronization upon reconnection.

5.5 Phase 5: User Interface Design

The mobile interface is designed for users with minimal technical experience. It features large touch-friendly buttons, color-coded risk indicators (green/yellow/red), minimal text with clear labels, and support for both English and French (with future Pidgin support). Key screens include

Login/Registration, Dashboard (sales, profit, trust score), Payment Verification, Sales/Expense Entry, Financial Summary, and AI Insights/Alerts.

5.6 Phase 6: Testing & Evaluation

End-to-end testing uses simulated merchant scenarios covering the full workflow: payment verification, transaction tracking, summary generation, and trust scoring. Evaluation metrics include OCR extraction success rate, suspicious-payment detection accuracy, false-positive and false-negative rates, and trust-score consistency across similar merchant histories.

5.7 Phase 7: Deployment & Future Roadmap

The MVP is deployed targeting Android devices in Cameroon. The future roadmap includes integration with mobile money APIs (MTN, Orange), partnerships with financial institutions for trust score utilization, SMS-based transaction verification, advanced analytics, expansion to other African regions, and Pidgin language support.

6. Estimated Implementation Budget

This section provides a detailed breakdown of the estimated costs required to develop, test, and deploy the MboaTrust AI MVP within the framework of the ACIAI 2026 competition. All costs are expressed in FCFA (Franc CFA).

The budget reflects realistic implementation expenses for a student team building an AI-powered mobile platform. It covers cloud infrastructure, hardware, AI development resources, field research, and team operational support over an estimated 6-month development cycle. All coding, design, and AI model development work is performed directly by the team members.

6.1 Cloud Infrastructure & Software Services

This category covers cloud hosting, database services, paid software tools, and platform fees required to build, deploy, and operate the MVP for 12 months.

Table 3: Cloud Infrastructure & Software Services Costs

Item	Description	Cost (FCFA)
Cloud Server Hosting	Backend hosting on Firebase Blaze plan or AWS Lightsail for API server, AI inference, and data processing - 12 months.	240,000
Managed Database	Cloud-hosted PostgreSQL or Firestore database with automated backups - 12 months.	120,000
Cloud Storage	Object storage (Firebase Storage or AWS S3) for payment screenshots, processed images, and AI model files - 12 months.	60,000
Domain Name & SSL	Domain registration (.com or .cm) for 1 year. SSL via Let's Encrypt (free).	15,000
Google Play Developer	One-time registration fee (\$25) to publish the Android app on the Play Store.	16,000
CI/CD & DevOps Tools	GitHub Pro or GitLab for version control, CI/CD pipelines, and project management - 6 months.	35,000
UI/UX Design Tools	Figma Pro subscription for UI wireframing, prototyping, and collaborative design - 6 months.	55,000
Monitoring & Analytics	Application monitoring (Firebase Crashlytics, Analytics) and error tracking - 12 months.	25,000
Subtotal - Cloud & Software Services		566,000

6.2 Hardware & Testing Devices

This category covers physical devices needed for development, quality assurance, and field testing. The app must be tested on devices representative of the target users (low to mid-range Android phones common among informal merchants in Cameroon).

Table 4: Hardware & Testing Devices Costs

Item	Description	Cost (FCFA)
Low-End Android Phone	Budget Android device (Tecno Pop, Itel) representing the minimum spec of target users - for QA testing.	40,000
Mid-Range Android Phone	Mid-tier device (Samsung Galaxy A series or Xiaomi Redmi) for performance testing and UI validation.	90,000
SIM Cards & Airtime	2 prepaid SIM cards with mobile money accounts (MTN + Orange) for testing real payment screenshot workflows.	20,000
Portable Power Bank	Power bank for field testing sessions in markets without reliable electricity.	15,000
USB Cables & Accessories	USB debugging cables, OTG adapters, and phone stands for development and testing.	10,000
Subtotal - Hardware & Testing		175,000

6.3 AI Development & Data Resources

This category covers the computational resources, tools, and data acquisition costs specific to developing the five AI/ML modules (OCR, fraud detection, anomaly detection, financial behavior analysis, and trust scoring). Open-source frameworks (TensorFlow, PyTorch, Tesseract) are used at no cost; this budget covers the paid compute and data resources needed.

Table 5: AI Development & Data Resources Costs

Item	Description	Cost (FCFA)
Google Colab Pro+	GPU-accelerated cloud environment (A100/V100 GPUs) for training OCR, fraud detection, and anomaly detection models - 6 months.	210,000
Cloud GPU Burst (AWS/GCP)	On-demand GPU instances for intensive model training sessions and hyperparameter tuning beyond Colab limits.	300,000
Synthetic Data Generation	Compute costs for generating simulated payment screenshots, transaction records, and merchant histories at scale.	80,000
Data Annotation Tools	Label Studio or similar open-source tool hosted on cloud for team-based annotation of training data (screenshots, fraud labels).	45,000

Pre-trained Model Access	Hugging Face Pro subscription for accessing optimized OCR and NLP model weights for fine-tuning.	65,000
Model Storage & Versioning	MLflow or Weights & Biases for experiment tracking, model versioning, and performance comparison - 6 months.	50,000
Subtotal - AI Development & Data		750,000

6.4 Research, Field Work & User Testing

This category covers costs related to user research in the field, usability testing with actual merchants, documentation, and competition deliverables. Direct engagement with bayam-sellam traders is essential for validating the product design and ensuring real-world usability.

Table 6: Research, Field Work & User Testing Costs

Item	Description	Cost (FCFA)
Transport (Field Research)	Local transport to markets in Douala and Yaoundé for user interviews, observation sessions, and usability testing - multiple trips over 3 months.	120,000
Merchant Participation	Small compensation for merchants participating in user testing sessions (time away from their stalls).	100,000
Refreshments & Logistics	Water, snacks, and venue logistics for organized user testing and feedback sessions.	30,000
Airtime & Communication	Phone credit for coordinating with merchants, scheduling testing sessions, and collecting remote feedback.	25,000
Printing & Documentation	Printing of SRS documents, questionnaires, consent forms, competition reports, and presentation materials.	40,000
Photography & Recording	Memory cards and minor equipment for documenting field visits, user testing sessions, and demo recordings.	20,000
Subtotal - Research & Field Work		335,000

6.5 Operations & Team Support

This category covers ongoing operational costs that sustain the development team over the 6-month project lifecycle, including internet, workspace, electricity, and team coordination expenses. While team members are not salaried, basic operational support ensures consistent productivity.

Table 7: Operations & Team Support Costs

Item	Description	Cost (FCFA)
Internet Data Bundles	High-speed mobile data and/or broadband subscriptions for 2 team members - 6 months. Essential for cloud development, model training, code pushing, and deployment.	210,000
Electricity & Power	Contribution toward electricity costs for laptop and device charging during development - 6 months.	60,000
Co-working Space	Periodic co-working space or lab rental for focused collaborative development sessions and sprint reviews.	90,000
Team Coordination	Transport for team meetings, whiteboard sessions, and sprint planning when not co-located.	50,000
Competition Travel	Transport and accommodation for attending ACIAI 2026 competition events, presentations, and demos.	250,000
Presentation Materials	Poster printing, demo video production, pitch deck design, and branded materials for competition presentation.	80,000
Stationery & Supplies	Notebooks, markers, sticky notes for brainstorming, sprint planning boards, and documentation.	15,000
Subtotal - Operations & Team Support		755,000

6.6 Consolidated Budget Summary

The table below consolidates all cost categories into an overall project budget, including a 10% contingency reserve for unforeseen expenses and scope adjustments during the development cycle.

Table 8: Consolidated Budget Summary

Budget Category	Cost (FCFA)
Cloud Infrastructure & Software Services (Section 6.1)	566,000
Hardware & Testing Devices (Section 6.2)	175,000
AI Development & Data Resources (Section 6.3)	750,000
Research, Field Work & User Testing (Section 6.4)	335,000
Operations & Team Support (Section 6.5)	755,000
Subtotal before Contingency	2,581,000

Contingency Reserve (10%)	258,100
TOTAL ESTIMATED BUDGET	2,839,100 FCFA

Currency Equivalents (approximate at current exchange rates):

- 2,839,100 FCFA $\approx$ 4,330 EUR
- 2,839,100 FCFA $\approx$ 4,730 USD

Budget Assumptions & Notes

- All development work (coding, AI model training, UI/UX design, testing, and integration) is performed directly by the team members. No external developer salaries or freelancer fees are included.
- Team members use their own personal laptops for development; no computer purchase is budgeted.
- The project uses open-source frameworks and tools at no cost: Python, TensorFlow/PyTorch, Tesseract OCR, Android Studio, Git, and VS Code.
- AI model training leverages Google Colab Pro+ and on-demand cloud GPU instances for intensive training runs, ensuring high-quality model performance.
- Synthetic and simulated datasets are created by the team. No proprietary dataset purchases are required.
- Cloud infrastructure costs assume moderate usage appropriate for an MVP serving early adopter merchants. Costs would scale with user growth in a post-competition deployment.
- Competition travel and presentation costs are included to cover ACIAI 2026 event participation.
- This budget covers the full MVP development cycle (approximately 6 months of development + 12 months of infrastructure operation). A full commercial-scale deployment would require additional investment in dedicated servers, security audits, marketing, and team compensation.